



Sample Problems for the Final Exam

CSE 373: Design and Analysis of Algorithms

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1. Dijkstra's Algorithm (Shortest Path)

Conceptual Questions

D1. Why does Dijkstra's algorithm require all edge weights to be non-negative?

Simulation Questions

DS1. A navigation system is being designed for a city's road network. Each intersection is represented as a node, and each road segment between intersections has a **non-negative travel time** (in minutes).

The city map is modeled as the following **weighted graph**, where nodes represent intersections and edge weights represent travel times:

Edges with travel times (minutes):

$A \leftrightarrow B : 4$
 $A \leftrightarrow C : 2$
 $B \leftrightarrow C : 5$
 $B \leftrightarrow D : 10$
 $C \leftrightarrow D : 4$
 $C \leftrightarrow E : 3$
 $D \leftrightarrow E : 1$

The navigation system wants to compute the shortest travel time from a central hub, intersection **A**, to all other intersections.

Task:

Simulate **Dijkstra's algorithm** starting from source node **A**. At each step, clearly show:

- Which node is extracted as the minimum-distance unvisited node
- How the distance estimates to its neighboring nodes are updated
- The contents of the distance array after each extraction

Continue the simulation until the shortest distances from node **A** to all other nodes have been finalized.

2. Prim's Algorithm (Minimum Spanning Tree)

Conceptual Questions

P1. Explain the intuition behind Prim's algorithm using a "growing tree" analogy.

P2. What role does the priority queue play in Prim's algorithm?

Simulation Questions

PS1. Using the graph provided, simulate Prim's algorithm starting from vertex A. At each step, show:

- chosen edge
- updated MST set
- priority queue contents

PS2. For the following weighted edges, construct the MST:

$$(A, B, 4), (A, C, 2), (C, E, 3), (E, F, 5), (B, D, 10), (E, D, 4)$$

PS3. A city is planning to build a new **fiber-optic network** to connect several important facilities. Each facility is represented as a node, and the cost (in millions of taka) to directly connect two facilities is represented by a weighted, undirected edge.

The goal is to connect **all facilities with minimum total cost**, ensuring that the network remains connected without forming cycles.

The facilities and connection costs are given below:

Facility 1	Facility 2	Cost
A	B	4
A	C	3
B	C	1
B	D	2
C	D	4
C	E	5
D	E	2

Assume the network is modeled as a connected, undirected, weighted graph.

Task:

Simulate **Prim's algorithm** starting from node **A**.

- (a) At each step, specify:
- the edge selected,
 - the set of vertices included in the MST so far,
 - the current total cost of the MST.
- (b) Continue the simulation until the Minimum Spanning Tree is complete.
- (c) Write the final set of edges in the MST and the total cost.

3. Greedy Algorithms

Conceptual Questions

G1. Explain with an example why a greedy choice may fail for the 0/1 Knapsack problem.

Simulation Questions

A disaster relief organization has a helicopter that can carry at most **120 kg** of supplies. There are several types of relief items available, each with a known **weight** and **importance score**. The organization is allowed to take *fractions* of items if necessary.

Item	Weight (kg)	Importance Score
Medical Kits	30	90
Food Packs	40	100
Water Bottles	50	80
Blankets	60	120

Tasks:

- Explain why this problem is suitable to be solved using a **greedy strategy**.
- Identify the greedy choice criterion that should be used.
- Apply the greedy algorithm step by step to determine which items (or fractions of items) should be loaded onto the helicopter to maximize the total importance score.
- State the final total importance score obtained.

4. Dynamic Programming: Longest Common Subsequence (LCS)

Conceptual Questions

- L1.** State the recursive formula for the LCS problem and explain it in simple words.
- L2.** What are overlapping subproblems in the context of LCS? Give an example from the DP matrix.
- L3.** Why does LCS use a 2D DP table while Fibonacci requires only 1D?

Simulation Questions

LS1. Compute the LCS table for the strings:

$$X = \text{"ABCD"}, \quad Y = \text{"ACBD"}$$

and backtrack to find the LCS.

- LS2.** Identify all subproblems of the form $LCS(i, j)$ that are solved more than once in the recursive (non-DP) approach.
- LS3.** In bioinformatics, scientists often compare two DNA sequences to find similarities. One common measure of similarity is the **Longest Common Subsequence (LCS)**. Consider the following two sequences:

$$X = \text{AGGTAB}, \quad Y = \text{GXTXAYB}$$

Task:

- Define the recurrence relation used to compute the LCS length.
- Construct the DP table step by step and fill in all values.
- Identify the length of the LCS and one valid LCS sequence.

5. Shortest Path vs MST (Conceptual Comparison)

- C1.** Give two key differences between Dijkstra's and Prim's algorithm.
- C2.** Explain why adding a constant value to all edges does not change the MST but may change shortest paths.